physics

IMPACT FACTOR 1.5

CITESCORE 3.0

Editorial

New Advances in Quantum Geometry

Shi-Dong Liang, Tiberiu Harko and Matthew J. Lake

Special Issue

New Advances in Quantum Geometry

Edited by

Prof. Dr. Shi-Dong Liang, Prof. Dr. Tiberiu Harko and Dr. Matthew J. Lake

MDPI

https://doi.org/10.3390/physics5030045

physics

MDPI

Editorial

New Advances in Quantum Geometry

Shi-Dong Liang [1,2,*], Tiberiu Harko [3,4,*] and Matthew J. Lake [1,3,5,6,7,*]

1 School of Physics, Sun Yat-sen University, Guangzhou 510275, China
2 State Key Laboratory of Optoelectronic Material and Technology, Guangdong Province Key Laboratory of Display Material and Technology, Sun Yat-sen University, Guangzhou 510275, China
3 Department of Physics, Babeș-Bolyai University, Mihail Kogălniceanu Street 1, 400084 Cluj-Napoca, Romania
4 Department of Theoretical Physics, National Institute of Physics and Nuclear Engineering (IFIN-HH), 077125 Bucharest, Romania
5 National Astronomical Research Institute of Thailand, 260 Moo 4, T. Donkaew, A. Maerim, Chiang Mai 50180, Thailand
6 Department of Physics and Materials Science, Faculty of Science, Chiang Mai University, 239 Huaykaew Road, T. Suthep, A. Muang, Chiang Mai 50200, Thailand
7 Office of Research Administration, Chiang Mai University, 239 Huaykaew Road, T. Suthep, A. Muang, Chiang Mai 50200, Thailand
* Correspondence: stslsd@mail.sysu.edu.cn (S.-D.L.); tiberiu.harko@aira.astro.ro (T.H.); matthewjlake@narit.or.th (M.J.L.)

1. Introduction

Presently, we are in a period of rapid and intensive changes in our understanding of the gravitational interaction, triggered by the important observational findings of the late 1990s. We have witnessed the emergence of new cosmological paradigms and a better understanding of black hole properties, as well as a tremendous increase in the precision of observational data. However, in the understanding of one of the most fundamental questions of present day physics, the nature of quantum gravity, many unsolved questions still remain. Can gravity be quantized at all? Is the gravitational force a purely geometric effect, or is it field theoretic in nature? And, if gravity is pure geometry, what is the relation between geometry and the quantum? Is it possible to quantize geometry, to create a unified quantum geometric framework for gravity?

2. Scope and Aims of the Project

This Special Issue is focused on the fundamental question of quantum geometry, its various meanings, and its implications for the standard theoretical concepts in gravitation and cosmology. The Issue includes state-of-the-art research contributions in the following areas: the quantum geometry created by quantum matter [1], quantum metric fluctuations [2], cosmology in modified gravity models [3,4], de Sitter gauge theory [5], and matrix theory models of the gravitational interaction [6]. The nature of quantum configurations in phase space [7], momentum operators in intrinsically curved manifolds [8], uncertainty relations in the presence of a minimal length [9], generalized uncertainty black holes [10], and the effects of quantum gravity on mass scales at high energies [11] are also addressed. These fascinating topics, in which geometry, gravity and quantum mechanics are brought together, provide deeper insights into the unsolved mysteries of the gravitational interaction, at the smallest possible scales.

Review papers also play an essential role, both in synthesizing the available knowledge in a given field of science and in providing the key information required to understand the most advanced topics in contemporary research. Two reviews, one on noncommutativity in physics [12] and another on the Barbero–Immirzi parameter in loop quantum gravity [13], provide good introductions to these subjects, and overviews of some important recent results in these fascinating fields of basic research.

check for updates

Citation: Liang, S.-D.; Harko, T.; Lake, M.J. New Advances in Quantum Geometry. *Physics* **2023**, *5*, 688–689. https://doi.org/10.3390/physics5030045

Received: 26 June 2023
Accepted: 27 June 2023
Published: 30 June 2023

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

As the Guest Editors of this Special Issue, we hope that this collection will serve as a standard reference for initiating and continuing state-of-the-art research in the fundamental fields of quantum geometry, quantum gravity, and geometric theories of gravitation. Moreover, we hope that the Issue opens up some new perspectives on the quantum-geometric aspects of the gravitational field and its applications in astrophysics and cosmology. Our sincere thanks to all the authors who contributed to this volume, and without whom it would not have been possible, for their time and efforts.

Funding: This work was supported by the Grant of Scientific and Technological Projection of Guangdong Province (China), no. 2021A1515010036.

Conflicts of Interest: The authors declare no conflict of interest.

1. Ashtekar, A. Exploring quantum geometry created by quantum matter. *Physics* **2022**, *4*, 1384–1402. [CrossRef]
2. Haghani, Z.; Harko, T. Effects of quantum metric fluctuations on the cosmological evolution in Friedmann-Lemaitre-Robertson-Walker geometries. *Physics* **2021**, *3*, 689–714. [CrossRef]
3. Gadbail, G.N.; Mandal, S.; Sahoo, P.K. Parametrization of deceleration parameter in $f(Q)$ gravity. *Physics* **2022**, *4*, 1403–1412. [CrossRef]
4. Sofuoğlu, D.; Tiwari, R.K.; Abebe, A.; Alfedeel, A.H.A.; Hassan, E.I. The cosmology of a non-minimally coupled $f(R,T)$ gravitation. *Physics* **2022**, *4*, 1348–1358. [CrossRef]
5. Lu, J.-A. Cosmology of a polynomial model for de Sitter gauge theory sourced by a fluid. *Physics* **2022**, *4*, 1168–1179. [CrossRef]
6. Brahma, S.; Brandenberger, R.; Laliberte, S. BFSS matrix model cosmology: Progress and challenges. *Physics* **2023**, *5*, 1–10. [CrossRef]
7. Albuquerque, S.; Bezerra, V.B.; Lobo, I.P.; Macedo, G.; Morais, P.H.; Rodrigues, E.; Santos, L.C.N.; Varão, G. Quantum configuration and phase spaces: Finsler and Hamilton geometries. *Physics* **2023**, *5*, 90–115. [CrossRef]
8. Schürmann, T. On momentum operators given by Killing vectors whose integral curves are geodesics. *Physics* **2022**, *4*, 1440–1452. [CrossRef]
9. Rastegin, A.E. On Majorization uncertainty relations in the presence of a minimal length. *Physics* **2022**, *4*, 1413–1425. [CrossRef]
10. Lobos, N.J.L.S.; Pantig, R.C. Generalized extended uncertainty principle black holes: Shadow and lensing in the macro- and microscopic realms. *Physics* **2022**, *4*, 1318–1330. [CrossRef]
11. Singh, T.P. Why do elementary particles have such strange mass ratios?—The importance of quantum gravity at low energies. *Physics* **2022**, *4*, 948–969. [CrossRef]
12. Liang, S.-D.; Lake, M.J. An introduction to noncommutative physics. *Physics* **2023**, *5*, 436–460. [CrossRef]
13. Vyas, R.P.; Joshi, M.J. The Barbero–Immirzi parameter: An enigmatic parameter of loop quantum gravity. *Physics* **2022**, *4*, 1094–1116. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.